

Addressing Data Imbalance in Fetal Health Classification

Francisco Silva Cunha Campos

Faculty of Engineering & Faculty of Science, University of Porto

Porto, Portugal

up202108735@fe.up.pt

franciscosccampos@gmail.com

ABSTRACT

In this report, we will follow a case study on an imbalanced dataset: Fetal Health[3]. We aim to research how it is possible to enhance model's predictive power through different strategies, and how these strategies behave on different model paradigms.

1 INTRODUCTION

Reducing child mortality is an objective that is present in several of the Sustainable Development Goals[4] of the United Nations, and it is a crucial indicator of human progress, since it reflects the strength of healthcare, socioeconomic development, and equity.

Maternal mortality, closely related to child mortality, accounted for 295,000 deaths during and after pregnancy and childbirth in 2017. The vast majority of these deaths were largely preventable.

In this context, Cardiotocograms (CTGs)[1] are a simple and cost-effective tool for assessing fetal health. By emitting ultrasound pulses and analysing the response, CTGs provide a good framing on fetal heart rate (FHR), fetal movements, uterine contractions, and more, allowing timely interventions to prevent both child and maternal mortality.

Having this in mind, in this paper we aim to develop an AI system capable of correctly identifying the qualitative health of a fetus, in order to detect in a timely manner if intervention is needed, and to better allocate healthcare resources.

2 METHODOLOGY

In this section, we will discuss the methodology used to produce the results obtained in Section 3. We will pass through all the stages the research preparation: from dataset characteristics to the tools selected to fix the data imbalance issues on the Fetal Health dataset.

2.1 Dataset Characteristics

This dataset include data to classify the health of a fetus as Normal, Suspect or Pathological using CTG data. There are a total of 22 features in the dataset related to measurements done with CTGs (baseline value, fetal movement, accelerations, etc.). There are 2126 rows in the dataset.

2.2 Data Balancing Techniques

In order to choose the best data balancing technique, we decided to research the following oversampling methods:

- **Random Oversampling:** Randomly duplicate examples in the minority class.
- **SMOTE (Synthetic Minority Over-sampling Technique)[2]:** Generate synthetic examples in the minority class.

- **Borderline-SMOTE:** Generate synthetic examples in the minority class, but only for those that are close to the decision boundary.
- **ADASYN (Adaptive Synthetic Sampling Approach for Imbalanced Learning):** Generate synthetic examples in the minority class, but with a higher density of examples in the regions where the class distribution is sparse.
- **SMOTE-ENN (SMOTE + ENN):** Combine over- and under-sampling using SMOTE and Edited Nearest Neighbors.
- **SMOTE-TL (SMOTE + Tomek Links):** Combine over- and under-sampling using SMOTE and Tomek Links.

It was used one undersampling method: **Random Undersampling**, consisting of randomly deleting examples from the majority class.

2.3 Training Models

In order to study the performance of the balanced and unbalanced datasets on different model paradigms, it was decided to choose three training models of distinct nature:

- **Random Forest:** Robust ensemble method, handles imbalance well.
- **K-Nearest Neighbour:** Non-parametric, sensitive to class imbalance.
- **Logistic Regression:** Simple, interpretable linear model.

3 RESULTS AND DISCUSSION

3.1 Metrics Optimized

To select the best metrics, we must understand the consequences of classifying a fetus as Normal, Suspect, or Pathological. Misclassifying a Pathological (or even Suspicious) fetus as Normal (False Negative) can have fatal consequences, so minimizing False Negatives is essential. Conversely, we also want to avoid unnecessary resources on Normal fetuses misclassified as Pathological/Suspicious (False Positive), minimizing these as well (even though it's not the main priority).

Given the imbalanced nature of the dataset, **Accuracy** is not reliable, as it can be misleading with skewed class distributions. **Precision** is also insufficient since it doesn't account for False Negatives.

Recall is critical as it focuses on minimizing False Negatives. However, to balance both False Positives and False Negatives, **F1-Score** is an important performance metric to have in consideration.

In summary, we will focus on optimizing **Recall** and **F1-Score** for our model evaluations.

3.2 Imbalanced Dataset Results

In short, these were the main takeaways of the predictions made by the models trained on the imbalanced dataset:

- The Random Forest model had the average best performance in terms of Recall and F1 Score, with values of 0.873 and 0.906, respectively.
- Class 1 (Suspect) had the worst performance in terms of Recall and F1 Score, with values of Recall lower than 0.71 and F1 Score lower than 0.81.
- The biggest misclassification happened when predicting class 0 (Normal), when it was from class 1 (Suspect).

3.3 Balanced Dataset Results

It was tested how the different resampling methods affected the metrics we wanted to optimize.

Overall, these were the main takeaways we got for each of the models:

3.3.1 Random Forest.

- The best techniques was **RandomOverSampler** with a average Recall and F1-Score of 0.819 and 0.715, respectively.
- The worst technique was **SMOTEEN** with a average Recall and F1-Score of 0.731 and 0.540, respectively.

3.3.2 KNN.

- The best techniques were **RandomOverSampler** with a average Recall and
- The worst technique was **SMOTEENN** with a average Recall and F1-Score of 0.796 and 0.644, respectively.
- **ADASYN** and **SMOTE** were also strong contenders for this model.

3.3.3 Logistics Regression.

- We can consider that the best technique was **RandomUnderSampler** with a average Recall and F1-Score of 0.733 and 0.532.
- The worst overall technique was **ADASYN** with a average Recall of 0.675 and a F1-Score of 0.507.

3.4 Final Comparison

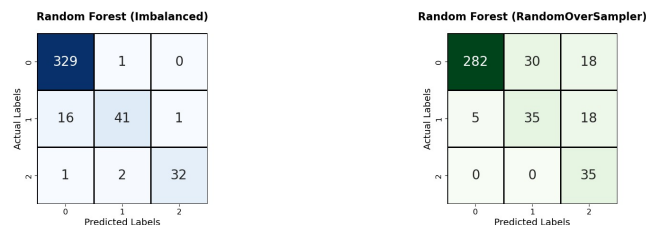


Figure 1: Confusion Matrix of Random Forest

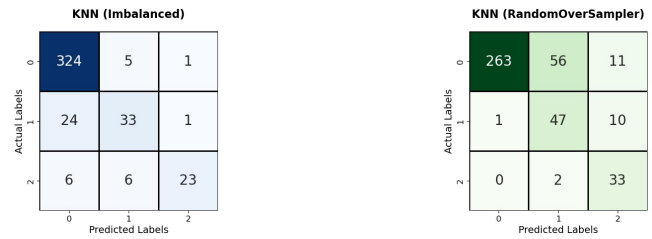


Figure 2: Confusion Matrix of KNN

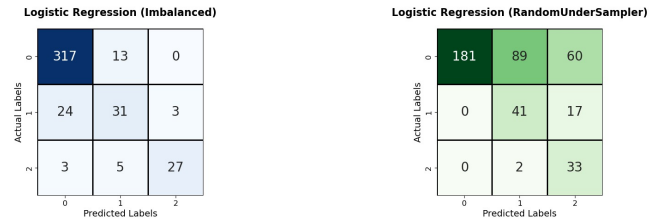


Figure 3: Confusion Matrix of Logistic Regression

For the **imbalanced dataset**, the Random Forest model had the best performance in terms of Recall and F1 Score, then the K-Nearest Neighbors model and finally the Logistic Regression model. However, when we **balanced the dataset**, the K-Nearest Neighbors model had the best performance in terms of Recall and F1 Score, then the Random Forest model and finally the Logistic Regression model.

4 CONCLUSIONS

In general, we can observe that there has been a shift from overall prediction in the Normal class (majority class) to the Pathological and Suspect classes (minority classes). This is because the models are now more balanced, and therefore they are not biased towards making predictions in the majority class.

We can also observe that has been a decrease in the F1 Score for all models, which is expected, because the models are now more balanced, and therefore they are not biased towards making predictions in the majority class (decreasing accuracy). Recall has also decreased, but that is because it is taking into account False Negatives from the other classes, and not only the majority class. Coming back to the initial analysis, we want to prevent a fetus that is Pathological/Suspicious from being classified as Normal. With this in mind, we can say that the models are now better at classifying the minority classes, which is what we wanted to achieve.

REFERENCES

- [1] D. Ayres de Campos et al. 2000. SisPorto 2.0: A Program for Automated Analysis of Cardiotocograms. *Journal of Maternal-Fetal Medicine* 5, 5 (2000), 311–318. [https://doi.org/10.1002/1520-6661\(200009/10\)9:5<311::AID-MFM12>3.0.CO;2-9](https://doi.org/10.1002/1520-6661(200009/10)9:5<311::AID-MFM12>3.0.CO;2-9)
- [2] G. E. A. Batista, R. C. Prati, and M. C. Monard. 2004. A Survey of Some Methods for Balancing Imbalanced Datasets. *ACM Computing Surveys (CSUR)* 44, 4 (2004), 1–39. <https://doi.org/10.1145/1035129.1035130>
- [3] Andrew MVD. 2025. Fetal Health Classification. <https://www.kaggle.com/datasets/andrewmvd/fetal-health-classification/data>
- [4] United Nations. 2020. Sustainable Development Goals. <https://sdgs.un.org/goals> Accessed: 2025-01-05.